THE MASTERMIND COACHING CLASSES

CBSE / RBSE • NURSERY – CLASS XII • JAIPUR

Road No. 5, VKIA, Jaipur — 9887258679

CLASS 10 — SCIENCE · NCERT-TYPE CHAPTER TEST SET 3

Chapter 9: **Light - Reflection and Refraction** | RBSE Pattern | Session 2026–27

Time: 1 Hour 30 Minutes Name: _____ Roll No.: _____ **Maximum Marks: 40**

General Instructions:

1. All questions are compulsory.
2. Q. 1–10 are Objective Type (1 mark each).
3. Q. 11–14 are Very Short Answer Type (2 marks each).
4. Q. 15–18 are Short Answer Type (3 marks each).
5. Q. 19–20 are Long Answer Type (5 marks each).
6. Internal choice is given in Q. 17 to 20.
7. Write the question number before each answer.
8. NCERT-type: questions in the style of the textbook examples and exercises.

SECTION — A (Objective Type Questions) 10 × 1 = 10

Choose the correct option and write it in your answer book.

1. A concave lens always forms 1

(a) a virtual, erect and diminished image **(b)** an inverted image

(c) a magnified image **(d)** a real image

2. The rear-view mirrors of vehicles are convex because they 1

(a) give a wider field of view and erect, diminished images

(b) give enlarged images

(c) are cheaper than plane mirrors

(d) form real images

3. The refractive index of water with respect to air is 4/3. The refractive index of air with respect to water is 1

(a) 4/3 **(b)** 3/4 **(c)** 12 **(d)** 1

4. The magnification produced by a lens is negative. The image is 1

(a) real and erect **(b)** virtual and erect

(c) virtual and inverted **(d)** real and inverted

5. A ray passing through the centre of curvature of a concave mirror, after reflection, 1

(a) retraces its path **(b)** goes parallel to the axis

(c) passes through the focus **(d)** passes through the pole

6. A coin kept in water appears raised because of 1

(a) scattering **(b)** dispersion

(c) refraction of light **(d)** reflection

7. The lens formula is 1

(a) 1/u - 1/v = 1/f **(b)** 1/v - 1/u = 1/f **(c)** v + u = f **(d)** 1/v + 1/u = 1/f

8. A ray parallel to the principal axis of a concave mirror, after reflection, passes through 1

(a) the centre of curvature **(b)** the pole

(c) the focus **(d)** infinity

9. The refractive index of a medium depends on 1

(a) the colour of the observer **(b)** the thickness of the medium

(c) the angle of incidence **(d)** the nature of the medium

10. A ray passing through the optical centre of a lens 1

(a) goes undeviated **(b)** is bent towards the axis

(c) is reflected **(d)** is bent away from the axis

SECTION — B (Very Short Answer Type) 4 × 2 = 8

Answer in about 20–30 words each.

11. Write the lens formula and the magnification formula. 2

12. Why do we prefer a convex mirror as a rear-view mirror? 2

13. What is meant by refractive index? Give its formula. 2

14. Why is a convex lens called a converging lens? 2

SECTION — C (Short Answer Type) 4 × 3 = 12

Answer in about 40–60 words each.

15. Find the speed of light in glass of refractive index 1.5. (c = 3 x 10^8 m/s) 3

16. State three uses of convex mirrors. 3

17. Two lenses of power +2 D and +3 D are placed in contact. Find the net power and the focal length. 3

OR

Light travels from air to water making an angle of incidence of 45 degrees. State the direction of bending and write the relation giving the angle of refraction. 3

18. How does a rectangular glass slab affect a ray of light? Explain the emergent ray. 3

OR

Where should an object be placed in front of a convex lens to get a real image of the same size? Find the image distance if f = 15 cm. 3

SECTION — D (Long Answer Type) 2 × 5 = 10

Answer in about 100–150 words each.

19. Describe the nature, position and size of images formed by a convex lens for different positions of the object. 5

OR

Draw ray-diagram reasoning to explain the formation of images by a concave lens and state its uses. 5

20. An object 5 cm in size is placed 30 cm from a convex lens of focal length 20 cm. Find the image distance, height and nature. 5

OR

Explain the sign conventions for spherical mirrors and lenses. 5

END OF QUESTION PAPER Total: 40 Marks